



**DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA**

Politecnico di Milano

Machine Learning (Code: 097683)

July 19, 2016

Name:	
Surname:	
Student ID:	
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Time: 2 hour 30 minutes

Prof. Marcello Restelli

Maximum Mark: 31

- The following exam is composed of **10 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided outside these spaces will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices** like laptops, smartphones, tablets and similar. As well, you are not allowed to bring with you any kind of note, book, written scheme or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Ex. 9	Ex. 10	Tot.
/ 5	/ 5	/ 5	/ 2	/ 2	/ 2	/ 2	/ 2	/ 3	/ 3	/ 31

Exercise 1 (5 points)

Describe the supervised learning technique denominated **Ridge regression** for regression problems.

Exercise 2 (5 points)

Describe the unsupervised learning technique denominated **K-means** for clustering problems.

Exercise 3 (5 points)

Describe the **policy iteration** technique for control problems on Markov Decision Processes.

Exercise 4 (2 point)

Categorize into either supervised, unsupervised or reinforcement learning the following Machine Learning problems:

1. Pricing 2nd hand cars;
2. Selecting the most profitable price for goods on a retail website;
3. Teach a robotic hand to grasp a glass of beer;
4. Extract interesting features from a picture.

Motivate your answers.

For each one of the previous problems suggest a method to solve it and motivate your choice.

1. **SUPERVISED:** by resorting to the characteristics of the car (e.g., production year, color, model) we might be able to predict the car price. In this case, since we have an ordering over the price, we are facing a regression problem and we might use a *linear regression* to solve it;
2. **REINFORCEMENT LEARNING:** if we consider a procedure where we propose a price to a client and she/he decides to accept or refuse the good and we assume an unlimited stock, we might resort to the MAB framework to select the best price among a given set. The technique we might use in this case is *UCB1*;
3. **REINFORCEMENT LEARNING:** in this case we might have a dataset with previously recorded grasping movements or we make the robot directly learn from experience. If we assume that the model of the process is unknown, in the former case we might choose *montecarlo estimation*, while in the latter one we might resort to *SARSA*;
4. **UNSUPERVISED:** generally the problem of extracting information from a dataset when we do not have any label/output is addressed by unsupervised learning. In this case, a technique like *PCA* might extract the directions in the data space where most of the variance resides.

Exercise 5 (2 point)

Consider $x, y \in \mathbb{R}^d$, $d \in \mathbb{N}$. Which ones of the following are similarity measures:

1. $k(x, y) = x^T y + (x^T y)^2$;
2. $k(x, y) = ck'(x, y) + k''(x, y) \cdot k'''(x, y)$, where k' , k'' and k''' are valid similarities in $\mathbb{R}^d$;
3. $k(x, y) = x^2 e^{-y}$ ($d = 1$);
4. $k(x, y) = \sqrt{(1 - \sin^2(x))} \sin(y - \pi/2)$, ($d = 1$).

Motivate your answers.

According to Mercer's theorem a similarity measure (valid to be used as kernel) should be continuous, symmetric and semi-definite positive. Thus:

1. TRUE: since $x^T y$ is a valid similarity measure, thus also $(x^T y)^2$ is a valid similarity measure. Moreover, if we sum them we still have a similarity measure;
2. TRUE/FALSE: if $c \geq 0$, we have a valid similarity since it is a sum and product of valid similarities, if $c < 0$ the form might be non positive definite;
3. FALSE: this is not symmetric, for instance $0 = k(0, 1) \neq k(1, 0) = 1$;
4. FALSE: with simple algebraic transformation we have $k(x, y) = -|\cos(x)|\cos(y)$, which is clearly not semi-definite positive.

Exercise 6 (2 point)

Tell if the following statement about Multi-Armed Bandit (MAB) are true or false and provide adequate motivations for your choices.

1. Usually MAB algorithms perform exploitative (greedy) choices to reduce the uncertainty about arms rewards;
2. A MAB algorithm which considers always the greedy choice (maximize the empirical expected reward) might get stuck in a suboptimal arm;
3. The MDP corresponding to a MAB setting has no states;
4. One of the options to solve the exploration/exploitation dilemma in the MAB setting is to consider some form of statistical upper bounds of the expected rewards of the arms.

1. FALSE: to reduce uncertainty one should perform *explorative* choices;
2. TRUE: it is the case of an arm a with the highest reward which gains zero reward the first time it has been pulled. In this case, by considering a greedy approach, the arm a will not be pulled ever again;
3. FALSE: it has a single state;
4. TRUE: in fact, this is exactly what UCB1 does.

Exercise 7 (2 point)

Comment the following statements about Markov Decision Processes (MDPs).

1. In a finite state MDP there are only Markovian, stationary and deterministic optimal policies;
2. Given a policy π for an MDP, we can compute the corresponding policy transition matrix P^π and the policy expected rewards R^π in each state;
3. Given the value function $V^{\pi^*}(s)$ corresponding to the optimal policy π^* on a specific MDP, we are able to execute the optimal policy π^* on the MDP;
4. There is a unique optimal value function $V^{\pi^*}(s)$ in an MDP.

1. FALSE: you are assured there is at least a Markovian, stationary and deterministic optimal policies, but there might be other not satisfying these properties;
2. TRUE: in the case we have the model of the MDP, i.e., we know the transition model P and the instantaneous rewards R , we are able to compute the corresponding transition probabilities $P^\pi = \sum_{a \in \mathcal{A}} \pi(a|s)P(s'|a, s)$ and the corresponding expected instantaneous rewards $R^\pi = \sum_{a \in \mathcal{A}} \pi(a|s)R(s, a)$;
3. FALSE: the fact that we know the optimal value function $V^{\pi^*}(s)$ does not imply that we know the policy used to obtain it;
4. TRUE: it is the unique fixed point of the bellman optimality operator (which is ensured to be unique since the operator is a contraction).

Exercise 8 (2 point)

Describe the **advantages** and **drawbacks** of the following choices about model selection in Machine Learning:

1. Increase the model complexity and fix the number of samples;
2. Fix the model complexity and the number of samples, but use ensemble techniques.

1. Increasing the model complexity should reduce the bias of the model (advantage), at the cost of increasing the variance (drawback).
2. Using ensemble technique might help in reducing the bias or/and the variance of the model (advantage) at the cost of increasing the computational time required to train the model (drawback).

Exercise 9 (3 points)

Consider an initial parameter vector $\mathbf{w}^{(0)} = [1 \ 0 \ 2]^T$ and a loss function of the form:

$$J(\mathbf{w}, x_i, t_i) = \frac{1}{4}(\mathbf{x}_i^T \mathbf{w} - t_i)^4$$

for a linear regression model $\hat{t}_i = \mathbf{x}_i^T \mathbf{w}$.

1. Derive the generic update formula for the parameter vector $\mathbf{w}^{(i)}$, given the datum $(\mathbf{x}_i, t_i)$, provided by the gradient descent.
2. Apply the derived formula to the datum $x_1 = [2 \ 1 \ 3]^T$, $t_1 = 7$ and a learning rate $\alpha = 0.5$.

$$\begin{aligned} \mathbf{w}^{(i)} &= \mathbf{w}^{(i-1)} - \alpha \left. \frac{\partial J(\mathbf{w}, x_i, t_i)}{\partial \mathbf{w}} \right|_{\mathbf{w}=\mathbf{w}^{(i-1)}} \\ &= \mathbf{w}^{(i-1)} - \alpha \frac{1}{4} \cdot 4(\mathbf{x}_i^T \mathbf{w}^{(i-1)} - t_i)^3 \mathbf{x}_i \\ &= \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - 0.5 \left([2 \ 1 \ 3] \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - 7 \right)^3 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - 0.5 \cdot 1 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 0.5 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.5 \\ 0.5 \end{bmatrix} \end{aligned}$$

Exercise 10 (3 points)

Suppose you collected data about players of an online Role-Playing-Game. For the i -th player you know the amount of hours played x_{i1} , the number of collected items x_{i2} , the distance covered (in km) during the game x_{i3} and if they completed the final quest t_i . We fit a logistic regression ($\hat{t}_i = \sigma(w_0 + \sum_j w_j x_{ij})$) to these data which predicts if a player completed the final quest t_i given the game statistics (x_{i1}, x_{i2}, x_{i3}) and produce the estimated coefficients: $w_0 = -5$, $w_1 = 0.05$, $w_2 = 0.1$ and $w_3 = 1$.

- Estimate the probability that a player who played 40h, collected 10 items and traveled for 1500m completed the final quest.
- Do you think that values of z in $\sigma(z)$ lower than -5 make sense in this problem? Why?
- According to the considered model, is it better to increase your weaponry (number of items) of 10 units or walk for 4000m?

z	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0
$\sigma(z)$	0.006	0.01	0.02	0.029	0.047	0.075	0.119	0.182	0.268	0.377	0.5
z	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5
$\sigma(z)$	0.622	0.731	0.817	0.88	0.924	0.953	0.971	0.982	0.989	0.993	0.996

If the value you need is not among the ones provided, approximate the $\sigma(z)$ value with the one for the nearest z .

The wanted probability is:

$$\hat{t}_i = \sigma(w_0 + \sum_j w_j x_{ij}) = \sigma(-5 + 40 \cdot 0.05 + 10 \cdot 0.1 + 1.5 \cdot 1) = \sigma(-0.5) = 0.377$$

where we converted the distance in kilometers.

It does not make sense to consider values of $z < -5$ since $x_{ij} > 0$, $\forall j$, thus in this specific problem we have that the lowest value for z is -5 (the value of the offset).

Finally, the increase in term of argument for having 10 more item is $\Delta z_1 = 10 \cdot 0.1 = 1$, while if we walk 4km more the increase is $\Delta z_2 = 4 \cdot 1 = 4$, thus:

$$\Delta z_1 < \Delta z_2 \Rightarrow \sigma(z + \Delta z_1) < \sigma(z + \Delta z_2)$$

for the monotonicity of $\sigma(\cdot)$. This implies that walking for 4000m increases the chance of completing the quest more than increasing the weaponry of 10 items.